PRODUCTION ARCHITECTURE SUBMISSION

September 2026

FreshSense.

Automated food freshness diagnostic platform powered by deep convolutional neural networks. Classifying produce degradation across five distinct biological stages in under 850 milliseconds on local CPU.



82%	11.5 MB	< 850ms	5 Stages	13 Species
MACRO F1 SCORE	MOBILENETV2 FOOTPRINT	LOCAL CPU LATENCY	BIOLOGICAL SPECTRUM	PRODUCE SCOPE

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MARKET NEED & INDUSTRY FLAW

Why Food Inspection Fails: The Binary Fallacy

Global agricultural wastage is compounded by subjective inspection and legacy machine learning systems that oversimplify quality into crude binary buckets.

01 / GLOBAL IMPACT

$1.3 Trillion Lost

Over one-third of all perishable food produced globally is discarded. Fruits and vegetables exhibit the highest spoilage rates, largely due to lack of objective early-stage quality evaluation.

33% TOTAL GLOBAL PRODUCE WASTE

02 / HUMAN INCONSISTENCY

Subjective Sensory Error

Manual sensory checking (smell, firmness touch, visual surface glance) varies widely across retail staff and home consumers, carrying an estimated 40% error rate in transitional decay onset.

40% MANUAL INSPECTION VARIANCE

03 / ALGORITHMIC FLAW

The Binary Fallacy

Existing AI models only categorize produce as Fresh or Rotten. They completely miss the transitional stages where produce is still edible, prime for culinary cooking, or markdown-ready.

80% LIFECYCLE STAGES MISSED

TAXONOMY & ACTIONABLE DIAGNOSTICS

The 5-Stage Freshness Spectrum

FreshSense mirrors real cellular and microbial degradation, transforming raw computer vision into immediate culinary and retail action.



STAGE 01

Fresh

Intact cellular turgor, vibrant uniform pigmentation, zero superficial pitting or fungal presence.

Action: Prime condition. Ideal for raw intake, salads, or extended refrigeration.



STAGE 02

Slightly Aged

Minor moisture transpiration, subtle dulling of skin luster, micro-softening at stem extremities.

Action: Safe for raw or cooked intake. Recommend consumption within 24–48 hours.



STAGE 03

Stale

Noticeable cellular collapse, surface wrinkling, moisture depletion, loss of natural aroma.

Action: Culinary quality degraded. Best cooked, stewed, or pureed; avoid raw salad use.



STAGE 04

Spoiled

Early microbial colonies, focal discoloration, softening pericarp, onset of off-odors.

Action: Unfit for consumption. High risk of foodborne toxins; discard immediately.



STAGE 05

Rotten

Advanced mycotoxin colonization, complete structural decomposition, pungent off-gassing.

Action: Hazardous bio-waste. Discard into sealed organic compost or disposal.

EMPIRICAL MACHINE LEARNING EVALUATION

Architecture Selection: Why MobileNetV2?

Trained on 6.41 GB across 13 species. Agglomerative clustering validated ground-truth degradation against 8 candidate vision backbones.

ARCHITECTURE	PARAMS	K-MEANS F1	AGGLOM ACC	MACRO F1
AlexNet	60.0M	0.70	0.84	0.83 (LFS Stub)
MobileNetV2	3.4M	0.75	0.82	0.82 SELECTED
DenseNet121	8.0M	0.74	0.83	0.82
Xception	22.8M	0.70	0.81	0.80
InceptionV3	23.8M	0.72	0.74	0.73
VGG16 / 19	138M	0.65	0.68	0.55
ResNet50	25.6M	0.23	0.42	0.34

KEY ARCHITECTURAL DECISION

Depthwise Separable Efficiency

MobileNetV2 achieves parity with heavy 60M–138M parameter models while using only 3.4M parameters. Depthwise separable convolutions factorize standard convolution into depthwise and pointwise layers, slashing compute by 8.5x.

PRODUCTION INTEGRITY

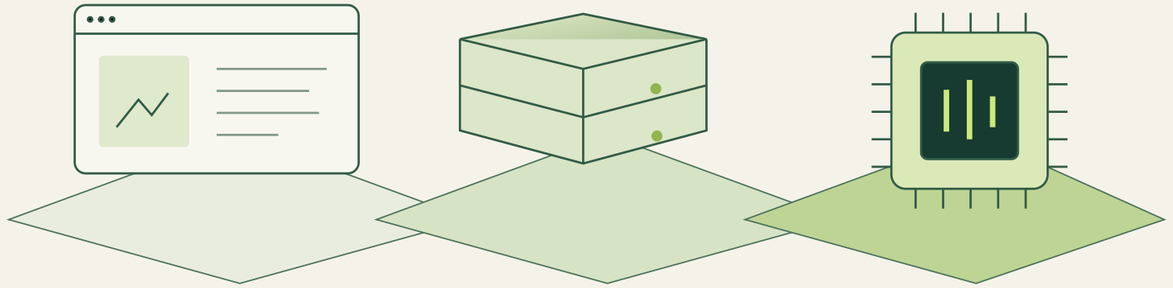
Intact Weights vs. Git LFS Stubs

While AlexNet showed a 0.01 higher metric in isolated notebooks, its weights were stranded as an unretrieved 134-byte Git LFS stub. MobileNetV2 ships an intact 11.5 MB binary with verified 0.82 F1 precision.

ROBUST EDGE-READY ARCHITECTURE

End-to-End Diagnostic Pipeline

From user image drag-and-drop to neural inference and persistent audit logging in under 850 milliseconds.



STEP 01

Client Ingestion

Drag & drop or file upload in browser. Client-side MIME validation (.jpg, .png, .webp) and instant preview.

Frontend • ES6

STEP 02

REST Gateway

Flask 3.1 microservice receives multipart payload, generates deterministic UUID, and streams to disk.

Flask • Python 3.14

STEP 03

Pillow Normalizer

Image standardized to RGB, bicubic resized to 128×128, and normalized to float32 [0.0, 1.0] tensor.

PIL 12.3 • NumPy

STEP 04

MobileNetV2 Engine

Keras 3 NumPy backend runs forward pass with verbose=0. Windows CRT Errno 22 patch eliminates I/O stalls.

Keras 3 • <850ms

STEP 05

Audit Log & UI

Appends result to history.json. Returns JSON payload. Client animates probability distribution bars.

history.json • UTF-8

RESILIENCE GUARANTEE: Local CPU inference • Zero external API dependencies • Zero GPU cost • Thread-safe UTF-8 persistence SLA: ≤ 1,200ms round-trip

DESIGN ENGINEERING & CLIENT QUALITY

Figma-Grade Diagnostic Interface

Crafted with modern typography, zero bulky framework overhead, and instant visual explainability.



UI HIGHLIGHT 01

Frictionless Upload

Interactive dropzone with visual drag-over feedback, instant thumbnail rendering, and single-click image removal. Works smoothly across touch screens and desktop browsers.

Drag • Drop CLIENT PREVIEW



UI HIGHLIGHT 02

Explainable Bars

Rather than a blind output label, FreshSense renders animated horizontal distribution bars showing the full probability breakdown across all five biological stages.

5 Classes FULL SOFTMAX VECTOR



UI HIGHLIGHT 03

Persistent Scan History

Chronological audit log rendering thumbnail previews, timestamps, and classified badges. Grocers and auditors can review inspection trends and flush records with one click.

history.json LOCAL AUDIT LOG

ENGINEERING TEAM • PROJECT DELIVERABLES

Cross-Functional Collaboration

Delivering an end-to-end computer vision product from data science research to production deployment.

MF

GROUP LEADER (GL)

Muhammad Faseeh

Project Strategy, AI Pipeline Integration, Backend Flask Architecture & Execution Reliability.

UI

TEAM MEMBER

Umamah Ibreeq Zafar

Computer Vision Research, Dataset Preparation, Unsupervised Clustering & Benchmark Evaluation.

AN

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Anumta Nadeem

UI/UX Design Engineering, CSS Design Systems, Interactive Animations & Client Testing.

ZA

TEAM MEMBER

Zafar Aman Khattak

System Architecture, Diagnostic Audit Logging, Windows CRT Resilience & Quality Assurance.

DEMO RUN COMMAND:

```
cd backend && python app.py
```

→ Open http://127.0.0.1:5000

Thank You • Questions & Evaluation